

Name:

Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



YEAR 11 PRELIMINARY COURSE

Extension 1 Mathematics

Assessment 2

July 2014

TIME ALLOWED: 75 minutes

Instructions:

- ***Start each question on a new page.***
- Write your name and class at the top of this page, and on your answer booklet.
- Write in blue or black pen only.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated within each question are a guide only and may be varied at the time of marking.
- It is suggested that you spend no more than 7 minutes on Section A.
- Approved calculators may be used.

SECTION A: (5 Marks)

Answers to these multiple choice should be completed on the multiple choice answer sheet supplied with your answer booklet.

All questions are worth 1 mark

1	$\frac{d}{dx} \left(\frac{5}{\sqrt{x}} \right) =$ A. $\frac{5x\sqrt{x}}{2}$ B. $\frac{5}{2x\sqrt{x}}$ C. $\frac{-5x\sqrt{x}}{2}$ D. $\frac{-5}{2x\sqrt{x}}$
2	$\sin(-120^\circ) =$ A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. $\frac{-\sqrt{3}}{2}$ D. $\frac{\sqrt{3}}{2}$
3	The acute angle between the line $x=3$ and the line $x - \sqrt{3}y + 2 = 0$ is: A. 60° B. 30° C. 90° D. 45°
4	If the endpoints of a diameter of the circle $(x - 2)^2 + (y + 1)^2 = 25$ are A $(-1, -5)$ and B (k, m) then the values of k and m are: A. $k = 5$ and $m = 3$ B. $k = 3$ and $m = 5$ C. $k = -4$ and $m = -9$ D. $k = -9$ and $m = -4$
5	Given that $\cos A = k$, $k > 0$, and $0^\circ \leq A \leq 90^\circ$, then $\sin 2A =$ A. $2\sqrt{1 - k^2}$ B. $2\sqrt{1 + k^2}$ C. $2k\sqrt{1 - k^2}$ D. $2k\sqrt{1 + k^2}$

SECTION B

(START EACH QUESTION ON A NEW PAGE)

QUESTION 6: (10 Marks)

	Marks
(a) Differentiate with respect to x :	
(i) $y = (3x^2 - 5)^3$	1
(ii) $y = \frac{x^3 - x^2 + 1}{x}$	1
(iii) $y = (x + 1)\sqrt{x + 1}$	2
(b) (i) Find the slope of the tangent to the curve $y = x^3 - x^2 - x + 1$ at the point where $x = 1$.	2
(ii) What does this imply about the x -axis and the curve at the point where $x = 1$?	1
(c) The lines $3x + 4y - 2 = 0$ and $3x + 4y + k = 0$ are 3 units apart. Find the two values of k .	3

QUESTION 7: (9 Marks) *(Start on a new page)*

		Marks
(a)	(i) Show that $\frac{1}{x+h} - \frac{1}{x} = \frac{-h}{x(x+h)}$	1
	(ii) Differentiate $f(x) = \frac{1}{x}$ using the method of First Principles.	3
(b)	Show that the equation of the tangent to the curve $y = \frac{x+2}{x-1}$ at the point where it crosses the x-axis is $x + 3y + 2 = 0$	3
(c)	Find the point P which divides the interval joining R(4, 3) to S(2, -1) externally in the ratio 3: 5	2

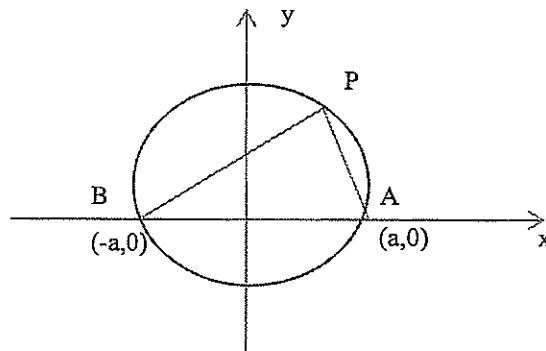
QUESTION 8: (9 Marks) (Start on a new page)

Marks

(a) Find $\lim_{h \rightarrow 0} \left\{ \frac{(5+h)^2 - 25}{h} \right\}$ 2

(b) Give the equation of the perpendicular bisector of the line which joins the points A (3, - 2) and B (5, 2). 3
Give your answer in general form.

(c) (i) Show that the point P ($a \cos \theta$, $a \sin \theta$) lies on the circle $x^2 + y^2 = a^2$ 1



(ii) Find the gradients of the lines BP and AP 2

(iii) Deduce that the line AP is at right angles to the line BP. 1

(Use ONLY the information in parts (i) and (ii). You are NOT to use the circle geometry proof related to the angle in a semi-circle)

QUESTION 9: (9 Marks) *(Start on a new page)*

			Marks
(a)	Show that $\tan(x + 45^\circ) = \frac{\sin x + \cos x}{\cos x - \sin x}$		2
(b)	(i) Find the equation of the normal to $y = x^3 - 2x^2 - 3x + 1$ at $P(2, -5)$.		2
	(ii) Show that there is another point on the curve where the normal to the curve is parallel to the normal at P. Find the co-ordinates this second point.		2
(c)	(i) Show that $\sin(A + B) + \sin(A - B) = 2\sin A \cos B$		1
	(ii) Hence, find the value of $\sin 75^\circ + \sin 15^\circ$		2

QUESTION 10: (9 Marks) *(Start on a new page)*

	Marks
(a) Find $\lim_{x \rightarrow \infty} \frac{2x^2+x}{3x^2-2}$	1
(b) (i) Show that the perpendicular distance of the point (4, 5) from the line $y = mx$ is: $d = \frac{ 4m-5 }{\sqrt{m^2+1}}$	1
(ii) If $y = mx$ is a tangent to the circle $(x - 4)^2 + (y - 5)^2 = 4$, explain why $\frac{ 4m-5 }{\sqrt{m^2+1}} = 2$	2
(c) (i) Show that $\cos 3A = 4\cos^3 A - 3\cos A$	2
(ii) Hence solve $4\cos^3 A - 3\cos A = \frac{1}{2}$ for $0^\circ \leq A \leq 180^\circ$	3

QUESTION 11: (9 Marks) (Start on a new page)

Marks

- (a) (i) Express $\cos\theta + \sqrt{3}\sin\theta$ in the form $R\cos(\theta - \alpha)$

2

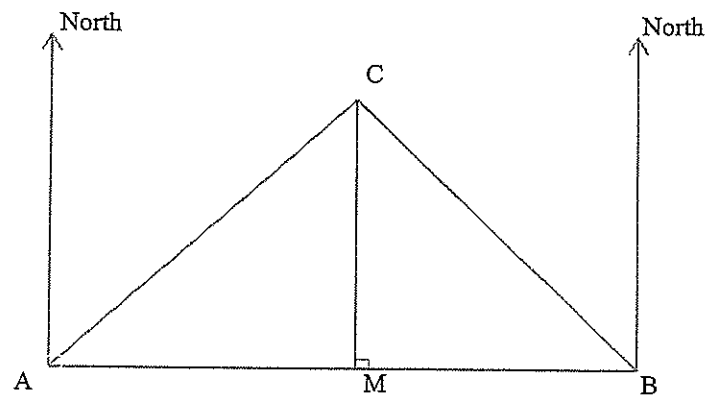
- (ii) Hence solve the equation

2

$$\cos\theta + \sqrt{3}\sin\theta = \sqrt{2} \quad \text{for } 0^\circ \leq \theta \leq 360^\circ$$

- (b) A surveyor stands at a point A and takes the bearing of a rock C, which he finds to be $N\alpha^\circ E$.
He walks due East 1000m to a second point B where he sees that C has a bearing of $N\beta^\circ W$.

He then moves to a third point, M, directly south of C.



- (i) Redraw the diagram above, and put on it all of the information contained in the question.

1

- (ii) Prove that from M the distance to the rock C is given by

4

$$\frac{1000\cos\alpha\cos\beta}{\sin(\alpha+\beta)}$$

YEAR 11 - SOLUTIONS - EXT1 JUNE 2014

SECTION A (1) D (2) C (3) A (4) A (5) C

SECTION B QUESTION 6:

(a) (i) $18x(3x^2 - 5)$

(ii) $2x - 1 - 1/x^2$

(iii) $\frac{dy}{dx} = \frac{d}{dx} (x+1)^{3/2}$
 $= \left\{ \frac{3/2 (x+1)^{1/2}}{3\sqrt{x+1}} \right\}$

2.
1 for getting
 $\frac{d}{dx} (x+1)^{3/2}$

(b) (i) $\frac{dy}{dx} = 3x^2 - 2x - 1$

At $x=1$ $\frac{dy}{dx} = 0$

(ii) the x -axis is a tangent.

(c) point on $3x + 4y - 2 = 0$ is $(0, +1/2)$

$p = 3 = \left| \frac{0 + 2 + k}{5} \right|$

$\therefore |k+2| = 15 \Rightarrow k = 13 \text{ or } k = -17$

← 1 each

QUESTION 7:

$$(a)(i) \quad \frac{1}{x+h} - \frac{1}{x} = \frac{x - (x+h)}{x(x+h)} \\ = -\frac{h}{x(x+h)}$$

1 MARK

$$(ii) \quad f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ = \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h} \\ = \lim_{h \rightarrow 0} \frac{-\frac{h}{x(x+h)}}{h} \\ = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} \\ = -\frac{1}{x^2}$$

← 1 for this line.

← 1 for writing part (i)

✓ 1 for answer.

Subtract 1 if the word "lim" is used incorrectly in any line.

$$(b) \quad \frac{dy}{dx} = \frac{(x-1)^{-1} - (x+2)^{-1}}{(x-1)^2} \\ = -\frac{3}{(x-1)^2}$$

← 1 MARK

At $y = 0$, $x = -2$

$$\therefore m_T = -3/9$$

$$\therefore m_T = -1/3$$

← 1 MARK

Equation is: $y = -1/3(x+2)$
 $x + 3y + 2 = 0$

← 1 MARK

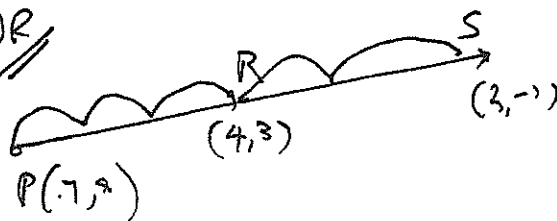
(c) $k_1 : k_2$
 $-3 : 5$

P is $\left(\frac{-3 \times 2 + 5 \times 4}{2}, \frac{3 \times 2 + 5 \times 4}{2} \right)$

$(4, 3) \rightarrow (2, -1) = (7, 9)$

2 MARKS

OR



QUESTION 8:

$$(a) \lim_{h \rightarrow 0} \left[\frac{25 + 10h + h^2 - 25}{h} \right]$$

$$= \lim_{h \rightarrow 0} (10 + h)$$

$$= 10$$

$$(b) \text{ Slope}_{AB} = \frac{4}{2} = 2$$

$$\text{Slope (perp)} = -\frac{1}{2}$$

$$\text{midpoint} = (4, 0)$$

$$y - 0 = -\frac{1}{2}(x - 4)$$

$$2y = -x + 4$$

$$x + 2y - 4 = 0$$

$$(c) (i) x^2 + y^2 = a^2 \cos^2 \theta + a^2 \sin^2 \theta = a^2$$

$$(ii) \text{ Slope}_{AP} = \frac{a \sin \theta}{a \cos \theta - a} = \frac{\sin \theta}{\cos \theta - 1}$$

$$\text{Slope}_{BP} = \frac{a \sin \theta}{a \cos \theta + a} = \frac{\sin \theta}{\cos \theta + 1}$$

$$(iii) m_{AP} \cdot m_{BP} = \frac{\sin^2 \theta}{\cos^2 \theta - 1} = \frac{1 - \cos^2 \theta}{\cos^2 \theta - 1} \left(\text{or } \frac{\sin^2 \theta}{-\sin^2 \theta} \right) = -1$$

$$\therefore \angle APB = 90^\circ$$

2 MARKS

lose 1 for omitting "lim" in line 2

← 1 MARK

← 1 MARK

← 1 for general form

← (1)

(1)

(1)

(1)

QUESTION 9:

$$(a) \tan(x+45^\circ) = \frac{\tan x + \tan 45}{1 - \tan x \tan 45}$$

$$= \frac{\tan x + 1}{1 - \tan x}$$

~~(b) (i)~~

$$= \frac{\sin x / \cos x + 1}{1 - \sin x / \cos x}$$

$$= \frac{\sin x + \cos x}{\cos x - \sin x}$$

1 for expansion and $\tan x = 1$

1 for simplification

$$(b) (i) \frac{dy}{dx} = 3x^2 - 4x - 3$$

$$\text{At } x=2 \quad m_T = 1 \Rightarrow m_N = -1$$

$$\therefore \text{Equation is } y+5 = -1(x-2)$$

$$x+y+3=0$$

(ii) If normals are parallel, so are the tangents

$$\therefore 3x^2 - 4x - 3 = 1$$

$$\therefore 3x^2 - 4x - 4 = 0$$

$$(3x+2)(x-2) = 0$$

$$\therefore x = 2 \quad \text{or} \quad x = -2/3$$

↑
found

$$\therefore y = -\frac{8}{27} - \frac{8}{9} + 2 + 1 = \frac{8}{27}$$

1 MARK

1 for realising this

1 for all this

1 for point

$$(c)(i) \sin(A+B) + \sin(A-B)$$

$$= \sin A \cos B + \cos A \sin B + \sin A \cos B - \cos A \sin B$$

$$= 2 \sin A \cos B$$

1 MARK

$$(ii) \text{ let } A = 45 \quad B = 30$$

$$\therefore \sin 75 + \sin 15 = 2 \sin 45 \cos 30$$

$$= 2 \times \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{3}}{\sqrt{2}} \quad \text{or} \quad \sqrt{\frac{3}{2}}$$

1 for this

①

QUESTION 10:

$$(a) \lim_{x \rightarrow \infty} \frac{2x^2 + x}{3x^2 - 2} = \lim_{x \rightarrow \infty} \frac{2 + \frac{1}{x}}{3 - \frac{2}{x^2}} = \frac{2}{3}$$

1 MARK (answer only)

(b)

$$(i) y = mx \Rightarrow mx - y = 0$$

$$d = \frac{|4m + 5(-1) + 0|}{\sqrt{m^2 + 1}} = \frac{|4m - 5|}{\sqrt{m^2 + 1}}$$

1 MARK

(ii) $y = mx$ is a tangent to the circle means the distance from the point of intersection to the centre of the circle is a radius. And is the shortest distance of the line from the origin i.e. d (above) = 2.

2 MARKS for a reasonable explanation.

$$\begin{aligned} (b)(i) \quad \cos 3A &= \cos(2A + A) \\ &= \cos 2A \cos A - \sin 2A \sin A \\ &= (2\cos^2 A - 1)\cos A - 2\sin^2 A \cos A \\ &= 2\cos^3 A - \cos A - 2(1 - \cos^2 A)\cos A \\ &= 4\cos^3 A - 3\cos A \end{aligned}$$

2 MARKS

$$(ii) 4\cos^3 A - 3\cos A = \frac{1}{2}$$

$$\text{means } \cos 3A = \frac{1}{2} \quad 0^\circ \leq 3A \leq 540^\circ$$

$$\therefore 3A = 60^\circ, 300^\circ, 420^\circ$$

$$\therefore A = 20^\circ, 100^\circ, 140^\circ$$

1 for realising this

2 MARKS

(Only 1 for only having 20°)

QUESTION 11:

(a) (i) $R = 2 \cos \theta + \sqrt{3} \sin \theta = 2 \left(\frac{1}{2} \cos \theta + \frac{\sqrt{3}}{2} \sin \theta \right)$

$= 2 \cos(\theta - \alpha)$

where $\cos \alpha = \frac{1}{2}$

$\Rightarrow \alpha = 60^\circ$

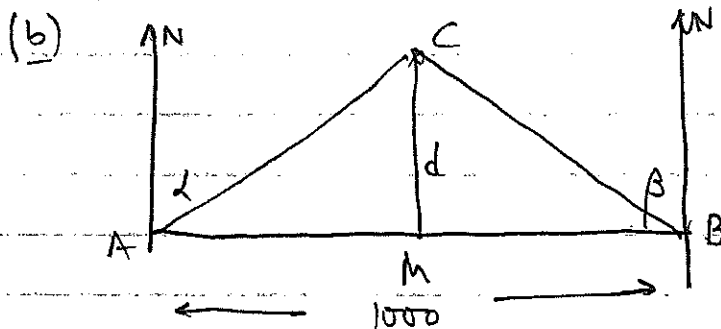
$\therefore \text{Exp}^N = 2 \cos(\alpha - 60^\circ)$

(ii) $2 \cos(\theta - 60^\circ) = \sqrt{2}$

$\therefore \cos(\theta - 60^\circ) = \frac{1}{\sqrt{2}}$

$\therefore \theta - 60^\circ = 45^\circ, 315^\circ$

$\therefore \theta = 105^\circ, 375^\circ, 15^\circ$



(ii) $\angle CAM = (90 - \alpha)^\circ$ $\angle CBM = (90 - \beta)^\circ$

In $\triangle CBM$, $\frac{d}{CB} = \sin(90 - \beta)$
 $= \cos \beta$

In $\triangle ABC$ $\frac{CB}{\sin(90 - \alpha)} = \frac{1000}{\sin(\alpha + \beta)}$ ← ①

$CB = \frac{1000 \cos(\alpha)}{\sin(\alpha + \beta)}$

$d = \frac{1000 \cos \alpha \cos \beta}{\sin(\alpha + \beta)}$

1 for R

1 for α .

②

1 for each value of θ

(note for 375°)

1 mark

← ① for finding CB

① for $\sin(90 - \beta)$
 $= \cos \beta$

and $\sin(90 - \alpha)$
 $= \cos \alpha$

1 mark